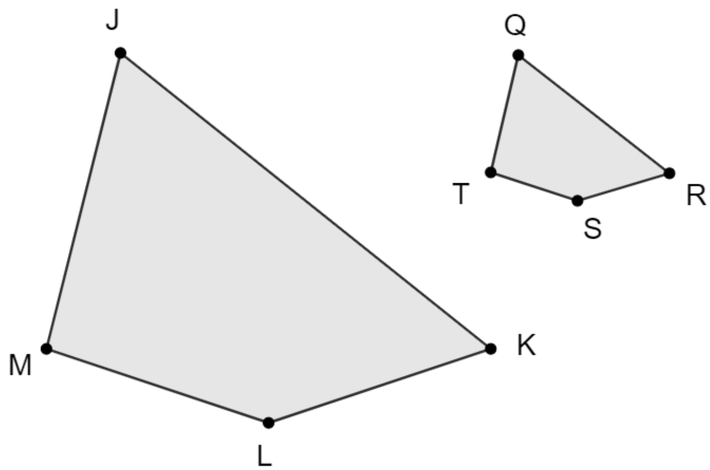
Geometry
Unit 6
Lesson 10 Homework

Name___Answer Key_______
Date____________________
Period__________________



Fill in the table given that $JKLM \sim QRST$.

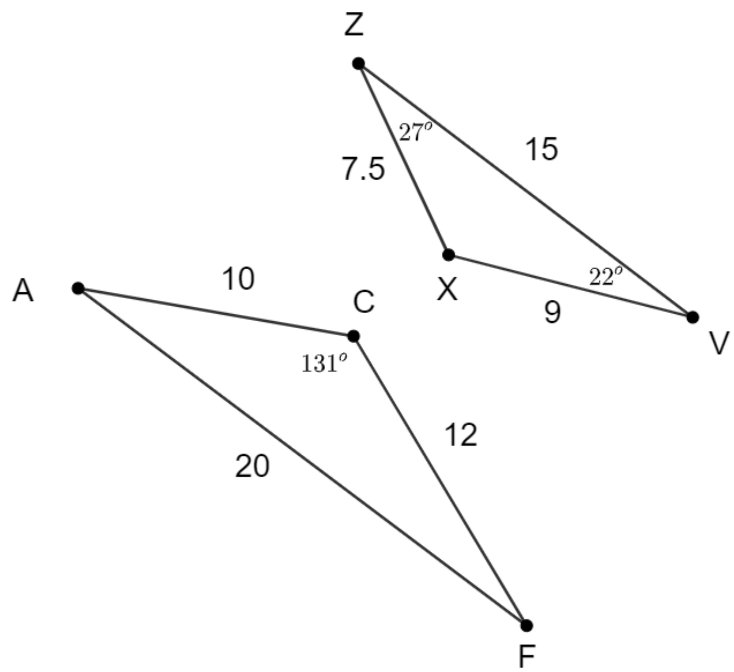
	Corresponding Angles	Corresponding Sides
1.	$\angle JML \cong \angle QTS$	$\frac{JM}{QT} = \frac{ML}{TS}$
2.	$\angle TQR \cong \angle MJK$	$\frac{LK}{SR} = \frac{JK}{QR}$
3.	$\angle TSR \cong \angle MLK$	$\frac{JK}{QR} = \frac{JM}{QT}$
4.	$\angle JKL \cong \angle QRS$	$\frac{ML}{TS} = \frac{LK}{SR}$

5. If $ML = 6.5$, $TS = 2.5$, and $QR = 5$, determine JK.

$$\frac{6.5}{2.5} = \frac{JK}{5}$$

$$JK = 13$$

$\triangle ACF$ and $\triangle VXZ$ are shown. $AC = 10$, $CF = 12$, $AF = 20$, $VX = 9$, $XZ = 7.5$, $ZV = 15$, $m\angle ACF = 131^{\text{o}}$, $m\angle XZV = 27^{\text{o}}$, and $m\angle ZVX = 22^{\text{o}}$.



6. What similarity theorem or postulate could be used to prove the triangles are similar?

Answers may vary. SAS ~ Theorem using $\frac{AC}{ZX} = \frac{CF}{XV}$ and $\angle C \cong \angle X$ or SSS.

7. Write a similarity statement for $\triangle ACF$ and $\triangle VXZ$.

$\triangle FCA \sim \triangle VXZ$

Complete the table.

	Angle Measures	Side Proportions
8.	$m\angle CAF =$ 27°	$\frac{AC}{ZX} = \frac{4}{3}$
9.	$m\angle CFA =$ 22°	$\frac{CF}{XV} = \frac{4}{3}$
10.	$m\angle ZXV =$ 131°	$\frac{FA}{VZ} = \frac{4}{3}$